Muffakham Jah College of Engineering and Technology

[Estd : 1980]
(Sultan-Ul-Uloom Education Society)
UGC Autonomous Institution
Approved by AICTE, Affiliated to Osmania University
Accredited by NAAC with A+Grade, NBA [Civil, Mech, ECE, CSE]
(Osmania University Approved Research Centres in Civil, CSE, ECE & Mech)

CSE (ARTIFICIAL INTELLIGENCE & MACHINE LEARNING) DEPARTMENT

B.E - [VII Semester] Major Project Abstract Sheet **BATCH: 2026-2027**

Project Title: KNEXUS – Knowledge Network for EXploration and Unified Services

Course Code: _____ **Branch:** _____ **Team No.** _____ **Team Size:** _____

TEAM MEMBERS DETAILS			PROJECT GUIDE DETAILS
Student Name & Sign	**Student Name & Sign**	**Student Name & Sign**	**Guide Name**
Roll No.: _____ **Mobile No.** _____	**Roll No.:** _____ **Mobile No.** _____	**Roll No.:** _____ **Mobile No.** _____	**Designation & Dept.**

Domain / Area:	**Artificial Intelligence and Machine Learning — Generative AI, Retrieval-Augmented Generation (RAG), Natural Language Processing, and Knowledge Management.**

Project Abstract

Organizations often store critical information across scattered documents, policies, standard operating procedures (SOPs), and technical manuals, making it difficult for employees to access accurate information efficiently. Existing AI assistants may generate unsupported answers and lack adequate access control and accountability when interacting with organizational systems. To address these challenges, **KNEXUS** is proposed as an AI-powered company knowledge platform that enables secure information retrieval, evidence-based question answering, and controlled task execution.

The system employs Retrieval-Augmented Generation (RAG) to process organizational documents, retrieve relevant information, and generate context-aware answers supported by source citations. It incorporates document uploading, text extraction, chunking, embedding generation, and semantic search. Role-Based Access Control (RBAC) and permission-aware retrieval ensure that users can access only information authorized for their organization and role. Additionally, the platform includes a controlled tool registry, human approval for designated actions, execution verification, and audit logging to ensure secure and transparent task execution. The system will be developed using React, TypeScript, Python FastAPI, PostgreSQL with pgvector, and locally hosted language models through Ollama.

The primary objective of KNEXUS is to develop a functional, modular, and cost-effective prototype that improves organizational knowledge accessibility while minimizing unsupported responses and unauthorized actions. The platform will be evaluated using retrieval relevance, citation accuracy, access-control testing, and controlled workflow execution. KNEXUS aims to establish a scalable foundation for intelligent organizational knowledge management, with future extensions for additional integrations and enterprise capabilities.

Signature of Guide		**Signature of HOD**	
Signature of Project Coordinator			